Calculus I Assignment 9

1. For the following problems, find the indicated integral AND check your answers by differentiation (show this !):

(a) $\int \frac{x^2+2x+1}{x^2}dx$

(b) $\int (x^3-2x^2)(\frac{1}{x}-5)dx$

(c) $\int \sqrt{t}(t^2-1)dt$

(d) $\int (e^y+1)^2 dy$

2. Find the general antiderivative:

(a) $\int 2\sec x \tan x dx$

(b) $\int x^2\sqrt{x^3+1}dx$

(c) $\int \frac{\cos x}{\sin x}dx$

(d) $\int 4xe^{2x^2+1}dx$

3. Use mathematical induction to prove that $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}$ for all integers $n \geq 1$.

Prove that $\int_a^b x^3 dx = \frac{b^4-a^4}{4}$

4. Compute the average value of the function $f(x) = x^2 + 2x$ on the interval $[0, 1]$.

5. A function f is said to be even if $f(-x) = f(x)$ for all x. A function f is said to be odd if $f(-x) = -f(x)$. Suppose that f is continuous for all x. Show that if f is even, then $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$. Also, if f is odd, then $\int_{-a}^{a} f(x)dx = 0$.

END